

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Tetrabutylammonium hydroxide, 40 wt.% solution in methanol  
 Cat No. : Q14163  
 Molecular Formula C16 H37 N O

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

Company Thermo Fisher Scientific India Pvt. Ltd  
 403-404, B-wing, Delphi,  
 Hiranandani Business Park,  
 Powai, Mumbai 4000076, INDIA

E-mail address laboratorysolution@thermofisher.com

### 1.4. Emergency telephone number

India Toll Free: 18 00 22 22 30  
 Chemtrec US: (800) 424-9300  
 Chemtrec EU: 001 (202) 483-7616

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

##### Physical hazards

Flammable liquids Category 2 (H225)

##### Health hazards

Acute oral toxicity Category 3 (H301)  
 Acute dermal toxicity Category 3 (H311)  
 Acute Inhalation Toxicity - Vapors Category 3 (H331)  
 Skin Corrosion/Irritation Category 1 B (H314)  
 Serious Eye Damage/Eye Irritation Category 1 (H318)  
 Specific target organ toxicity - (single exposure) Category 1 (H370)

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## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

## Hazard Statements

- H225 - Highly flammable liquid and vapor
- H314 - Causes severe skin burns and eye damage
- H370 - Causes damage to organs
- H301 + H311 + H331 - Toxic if swallowed, in contact with skin or if inhaled

## Precautionary Statements

- P280 - Wear protective gloves/protective clothing/eye protection/face protection
- P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
- P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
- P310 - Immediately call a POISON CENTER or doctor/physician
- P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor
- P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
- P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

## 2.3. Other hazards

- Toxic to terrestrial vertebrates
- This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                                  | CAS No    | EC No     | Weight % | CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567                      |
|--------------------------------------------|-----------|-----------|----------|--------------------------------------------------------------------------------------------------------------|
| Methyl alcohol                             | 67-56-1   | 200-659-6 | 60       | Flam. Liq. 2 (H225)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370) |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 2052-49-5 | 218-147-6 | 40       | Flam. Liq. 3 (H226)<br>Acute Tox. 4 (H302)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)                      |

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|  |  |  |  |                     |
|--|--|--|--|---------------------|
|  |  |  |  | Skin Sens. 1 (H317) |
|--|--|--|--|---------------------|

| Component      | Specific concentration limits (SCL's)                         | M-Factor | Component notes |
|----------------|---------------------------------------------------------------|----------|-----------------|
| Methyl alcohol | STOT Single Exp. 1 :: >= 10<br>STOT Single Exp. 2 :: 3 - < 10 | -        | -               |

| Components                   | Reach Registration Number |  |
|------------------------------|---------------------------|--|
| Methanol                     | 01-2119433307-44          |  |
| Tetrabutylammonium hydroxide | 01-2120231229-61          |  |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

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No information available.

## **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

## **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Thermal decomposition can lead to release of irritating gases and vapors.

## **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### **6.2. Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

## **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from heat, sparks and flame. Flammables area.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

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## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

| Component      | The United Kingdom                                                                                              | European Union                                               | Ireland                                                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Methyl alcohol | WEL - TWA: 200 ppm TWA;<br>266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL; 333 mg/m <sup>3</sup> STEL | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 200 ppm 8 hr.<br>TWA: 260 mg/m <sup>3</sup> 8 hr.<br>STEL: 600 ppm 15 min<br>STEL: 780 mg/m <sup>3</sup> 15 min<br>Skin |

#### Biological limit values

List source(s):

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                                         | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-------------------------------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Methyl alcohol<br>67-56-1 ( 60 )                                  |                              | DNEL = 20mg/kg<br>bw/day        |                                | DNEL = 20mg/kg<br>bw/day          |
| 1-Butanaminium,<br>N,N,N-tributyl-, hydroxide<br>2052-49-5 ( 40 ) |                              |                                 |                                | DNEL = 1.4mg/kg<br>bw/day         |

| Component                                                         | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Methyl alcohol<br>67-56-1 ( 60 )                                  | DNEL = 130mg/m <sup>3</sup>      | DNEL = 130mg/m <sup>3</sup>         | DNEL = 130mg/m <sup>3</sup>        | DNEL = 130mg/m <sup>3</sup>           |
| 1-Butanaminium,<br>N,N,N-tributyl-, hydroxide<br>2052-49-5 ( 40 ) |                                  |                                     |                                    | DNEL = 4.93mg/m <sup>3</sup>          |

#### Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                         | Fresh water     | Fresh water sediment            | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)           |
|-------------------------------------------------------------------|-----------------|---------------------------------|--------------------|------------------------------------|------------------------------|
| Methyl alcohol<br>67-56-1 ( 60 )                                  | PNEC = 20.8mg/L | PNEC = 77mg/kg<br>sediment dw   | PNEC = 1540mg/L    | PNEC = 100mg/L                     | PNEC = 100mg/kg<br>soil dw   |
| 1-Butanaminium,<br>N,N,N-tributyl-, hydroxide<br>2052-49-5 ( 40 ) | PNEC = 16.5µg/L | PNEC = 2.16mg/kg<br>sediment dw | PNEC = 0.165mg/L   | PNEC = 28.4mg/L                    | PNEC =<br>0.421mg/kg soil dw |

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| Component                                                         | Marine water    | Marine water sediment         | Marine water intermittent | Food chain | Air |
|-------------------------------------------------------------------|-----------------|-------------------------------|---------------------------|------------|-----|
| Methyl alcohol<br>67-56-1 ( 60 )                                  | PNEC = 2.08mg/L | PNEC = 7.7mg/kg sediment dw   |                           |            |     |
| 1-Butanaminium,<br>N,N,N-tributyl-, hydroxide<br>2052-49-5 ( 40 ) | PNEC = 1.65µg/L | PNEC = 0.216mg/kg sediment dw | PNEC = 16.5µg/L           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Neoprene       | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Inorganic gases and vapours filter Type B Grey Ammonia and organic ammonia derivatives filter Type K Green

### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

**Physical State** Liquid

**Appearance** Light yellow

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|                                            |                                       |                                          |
|--------------------------------------------|---------------------------------------|------------------------------------------|
| Odor                                       | Strong                                |                                          |
| Odor Threshold                             | No data available                     |                                          |
| Melting Point/Range                        | -98 °C / -144.4 °F                    |                                          |
| Softening Point                            | No data available                     |                                          |
| Boiling Point/Range                        | 65 °C / 149 °F                        |                                          |
| Flammability (liquid)                      | Highly flammable                      | On basis of test data                    |
| Flammability (solid,gas)                   | Not applicable                        | Liquid                                   |
| Explosion Limits                           | <b>Lower</b> 5.5<br><b>Upper</b> 36.5 |                                          |
| Flash Point                                | 12 °C / 53.6 °F                       | <b>Method</b> - No information available |
| Autoignition Temperature                   | 455 °C / 851 °F                       |                                          |
| Decomposition Temperature                  | No data available                     |                                          |
| pH                                         | No information available              |                                          |
| Viscosity                                  | No data available                     |                                          |
| Water Solubility                           | Soluble                               |                                          |
| Solubility in other solvents               | No information available              |                                          |
| Partition Coefficient (n-octanol/water)    |                                       |                                          |
| Component                                  | <b>log Pow</b>                        |                                          |
| Methyl alcohol                             | -0.74                                 |                                          |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 1.518                                 |                                          |
| Vapor Pressure                             | No data available                     |                                          |
| Density / Specific Gravity                 | 0.910                                 |                                          |
| Bulk Density                               | Not applicable                        | Liquid                                   |
| Vapor Density                              | No data available                     | (Air = 1.0)                              |
| Particle characteristics                   | Not applicable (liquid)               |                                          |

## 9.2. Other information

|                      |                                             |
|----------------------|---------------------------------------------|
| Molecular Formula    | C16 H37 N O                                 |
| Molecular Weight     | 259.46                                      |
| Explosive Properties | Vapors may form explosive mixtures with air |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Acids. Acid anhydrides. Acid chlorides. Metals. Reducing Agent.

### 10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 11: TOXICOLOGICAL INFORMATION

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## 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 3 |

### Toxicology data for the components

| Component                                  | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|--------------------------------------------|--------------------------------|-------------------------------|-------------------------------|
| Methyl alcohol                             | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 500 mg/kg (Rat)                | -                             | -                             |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

#### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

| Component                        | Test method                                                       | Test species | Study result    |
|----------------------------------|-------------------------------------------------------------------|--------------|-----------------|
| Methyl alcohol<br>67-56-1 ( 60 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

| Component                        | Test method             | Test species / Duration          | Study result              |
|----------------------------------|-------------------------|----------------------------------|---------------------------|
| Methyl alcohol<br>67-56-1 ( 60 ) | OECD Test Guideline 416 | Rat / Inhalation<br>2 Generation | NOAEC =<br>1.3 mg/l (air) |

(h) STOT-single exposure; Category 1

Results / Target organs Optic nerve, Central nervous system (CNS).

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be



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investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

| Component      | Freshwater Fish                            | Water Flea            | Freshwater Algae |
|----------------|--------------------------------------------|-----------------------|------------------|
| Methyl alcohol | Pimephales promelas: LC50 > 10000 mg/L 96h | EC50 > 10000 mg/L 24h |                  |

| Component      | Microtox                                                                        | M-Factor |
|----------------|---------------------------------------------------------------------------------|----------|
| Methyl alcohol | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min |          |

### 12.2. Persistence and degradability

#### Persistence

Readily biodegradable  
Persistence is unlikely, based on information available.

| Component                        | Degradability                  |
|----------------------------------|--------------------------------|
| Methyl alcohol<br>67-56-1 ( 60 ) | DT50 ~ 17.2d<br>>94% after 20d |

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component                                  | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------|---------|-------------------------------|
| Methyl alcohol                             | -0.74   | <10 dimensionless             |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 1.518   | No data available             |

### 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant

This product does not contain any known or suspected substance

#### Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

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|                                            |                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                   |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                   |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                                                            |
|-----------------------------------------|------------------------------------------------------------|
| <b>14.1. UN number</b>                  | UN3286                                                     |
| <b>14.2. UN proper shipping name</b>    | Flammable liquid, toxic, corrosive, n.o.s.                 |
| <b>Technical Shipping Name</b>          | Tetrabutylammonium hydroxide, 40 wt.% solution in methanol |
| <b>14.3. Transport hazard class(es)</b> | 3                                                          |
| <b>Subsidiary Hazard Class</b>          | 6.1, 8                                                     |
| <b>14.4. Packing group</b>              | II                                                         |

### ADR

|                                         |                                                            |
|-----------------------------------------|------------------------------------------------------------|
| <b>14.1. UN number</b>                  | UN3286                                                     |
| <b>14.2. UN proper shipping name</b>    | Flammable liquid, toxic, corrosive, n.o.s.                 |
| <b>Technical Shipping Name</b>          | Tetrabutylammonium hydroxide, 40 wt.% solution in methanol |
| <b>14.3. Transport hazard class(es)</b> | 3                                                          |
| <b>Subsidiary Hazard Class</b>          | 6.1, 8                                                     |
| <b>14.4. Packing group</b>              | II                                                         |

### IATA

|                                         |                                                            |
|-----------------------------------------|------------------------------------------------------------|
| <b>14.1. UN number</b>                  | UN3286                                                     |
| <b>14.2. UN proper shipping name</b>    | Flammable liquid, toxic, corrosive, n.o.s.                 |
| <b>Technical Shipping Name</b>          | Tetrabutylammonium hydroxide, 40 wt.% solution in methanol |
| <b>14.3. Transport hazard class(es)</b> | 3                                                          |
| <b>Subsidiary Hazard Class</b>          | 6.1, 8                                                     |
| <b>14.4. Packing group</b>              | II                                                         |

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| <b>14.5. Environmental hazards</b>                                   | No hazards identified            |
| <b>14.6. Special precautions for user</b>                            | No special precautions required. |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods   |

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

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## International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                  | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Methyl alcohol                             | 67-56-1   | 200-659-6 | -      | -   | X     | X    | KE-23193 | X    | X    |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 2052-49-5 | 218-147-6 | -      | -   | X     | X    | KE-34029 | X    | X    |

| Component                                  | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Methyl alcohol                             | 67-56-1   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 2052-49-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                                  | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------------------|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl alcohol                             | 67-56-1   | -                                                                   | Use restricted. See item 69.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 2052-49-5 | -                                                                   | -                                                                                                                                        | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                                  | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyl alcohol                             | 67-56-1   | 500 tonne                                                                                 | 5000 tonne                                                                               |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | 2052-49-5 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

# SAFETY DATA SHEET

Tetrabutylammonium hydroxide, 40 wt.% solution in methanol

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## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 2 (self classification)

| Component                                  | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                              |
|--------------------------------------------|---------------------------------------|------------------------------------------------------|
| Methyl alcohol                             | WGK 2                                 | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |
| 1-Butanaminium, N,N,N-tributyl-, hydroxide | WGK1                                  |                                                      |

| Component      | France - INRS (Tables of occupational diseases)      |
|----------------|------------------------------------------------------|
| Methyl alcohol | Tableaux des maladies professionnelles (TMP) - RG 84 |

| Component                        | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Methyl alcohol<br>67-56-1 ( 60 ) | Prohibited and Restricted Substances                                                                           | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H331 - Toxic if inhaled  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H370 - Causes damage to organs  
H225 - Highly flammable liquid and vapor  
H302 - Harmful if swallowed

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

# SAFETY DATA SHEET

Tetrabutylammonium hydroxide, 40 wt.% solution in methanol

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**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Creation Date** 04-Aug-2014

**Revision Date** 11-Oct-2023

**Revision Summary** Next Revision Date - Oct 2028.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

## Disclaimer

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**End of Safety Data Sheet**